

Approach - I

look,

$$\psi_H = \frac{1}{\sqrt{2}} (\psi_1 + \psi_2)$$

Original H- decompose to

Several H

$$H_0 = \sum_i |\psi_i\rangle \langle \psi_i|$$

$$H_0 = \sum_i c_i P_i$$

P_i → Pauli strings

Pauli operators

c_i → coefficient (real term).

~~Example~~

$$E(\theta) = \sum_i c_i \langle \psi | P_i | \psi \rangle$$

Why measure directly not possible?

$$\langle \psi | H | \psi \rangle$$

$$\psi^\dagger H \psi$$

$$E(\theta) = \langle \psi | H | \psi \rangle$$

Why did we convert $E(\theta)$ to $f(\theta)$?

→ As the Prob scale needs something that works like a probability.

$$\rightarrow 0 \leq f(\theta) \leq 1$$

But, $E(\theta)$ being energy can be

$$[-1, 1]$$

so we convert $E(\theta)$ to $f(\theta)$

$$f(\theta) = \frac{1 - E(\theta)}{2}$$

$$\Rightarrow f(\theta) = \frac{1 - \langle \psi | H | \psi \rangle}{2}$$

$$\Rightarrow 1 - f(\theta) = \frac{1 + E(\theta)}{2}$$

the probability scale, $\Rightarrow$

$$U_f |0\rangle |0\rangle = |0\rangle (\sqrt{f(\theta)} |1\rangle |\phi_1(\theta)\rangle + \sqrt{1-f(\theta)} |0\rangle |\phi_0(\theta)\rangle)$$

$$= |0\rangle (\sqrt{\frac{1-E(\theta)}{2}} |1\rangle |\phi_1(\theta)\rangle + \sqrt{\frac{1+E(\theta)}{2}} |0\rangle |\phi_0(\theta)\rangle)$$

$$\Rightarrow |0\rangle \left(\sqrt{\frac{1 - \langle \psi | H | \psi \rangle}{2}} |1\rangle |\phi_1(\theta)\rangle + \sqrt{\frac{1 + \langle \psi | H | \psi \rangle}{2}} |0\rangle |\phi_0(\theta)\rangle \right)$$

→ just works within qubit.

Q Prob oracle to phase oracle. ②

$$U_f |x\rangle |0\rangle = |0\rangle (\sqrt{1-f(x)} |0\rangle + \sqrt{f(x)} |1\rangle) \rightarrow \text{Prob oracle is simplified form with no varying orbits.}$$

Let's say

$$\sqrt{f(x)} = \sin(\alpha x) \rightarrow \sqrt{1-f(x)} = \cos(\alpha x)$$

$$\Rightarrow \text{so, } f(x) = \sin^2(\alpha x)$$

$$\Rightarrow U_f |x\rangle |0\rangle = |x\rangle (\cos(\alpha x) |0\rangle + \sin(\alpha x) |1\rangle)$$

Step ② Build a Grover's amplitude amplification operator.

→ why do we use this kind of amplitude amplification?

→ from Prob oracle we have αx as an hidden angle inside these operators.

→ To expose αx (hidden rotation angle) into an eigenphase we introduce this.

→ we introduce

$$Q_x = -A_x S_0 A_x^\dagger S_1$$

$$\text{Here, } A_x = U_f = \cos(\alpha x) |0\rangle\langle 0| + \sin(\alpha x) |1\rangle\langle 1|$$

$$S_0 = I - 2|0\rangle\langle 0|, S_1 = I - 2|1\rangle\langle 1| \quad \text{Reflection operators.}$$

Now Q_x acts like ~~rotation~~ rotation by $2\alpha x$

so, its eigenvalues are $e^{\pm 2i\alpha x}$

and then we use Phase Estimation to recover αx

and now we compute

$$f(x) = \sin^2(\alpha x)$$

ambitization. APPROACH - 2.

Prob to (2) (6)

... writing Has weighted sum

now, we have
 $f(x) = \sin^2(2x)$ & we compute Δx from
 $e^{i2\Delta x}$ $\rightarrow$ so now we can estimate $f(x)$

(3)

$\rightarrow$ Now come back to our initial state

$|x\rangle |f(x)\rangle |work\rangle \quad \Rightarrow R|y\rangle = e^{iy}|y\rangle$
 $\hookrightarrow$ Approximate. $\Rightarrow R|f(x)\rangle = e^{if(x)}|f(x)\rangle$
 $\Rightarrow |x\rangle |\tilde{f}(x)\rangle |work\rangle$

now apply controlled phase rotation.

$\rightarrow |\tilde{f}(x)\rangle = e^{i\tilde{f}(x)} |x\rangle |\tilde{f}(x)\rangle |work\rangle \rightarrow$ phase kick back.

now apply $U^\dagger$

$\rightarrow U(|x\rangle |0\rangle |0\rangle) = |x\rangle |f(x)\rangle |work\rangle \rightarrow$ we have this.

But now $\Rightarrow e^{if(x)} |x\rangle |f(x)\rangle |work\rangle$

if we apply $U^\dagger$
 $\rightarrow e^{if(x)} |x\rangle |0\rangle |0\rangle \Rightarrow U^\dagger$ doesn't apply on scalar phase
 so $f(x)$ & $f(x)$ & same with
~~work~~ $|work\rangle$ & $|work\rangle^\dagger$

now $|x\rangle |0\rangle = e^{if(x)} |x\rangle |0\rangle$

as extra $|0\rangle$ registers no longer have ~~anything~~ anything
 related to x or entangled with $|x\rangle$, we can remove them
 and say,

$|x\rangle = e^{if(x)} |x\rangle$

Note, (2)

At this point to

Note (2),
 But we need the preparation of select H blocks
 to prepare the block encoded H before
 measuring it to probability oracle.

Prepare the phase oracle,
 we don't explicitly need the
 probability oracle.

Now, we know $f(\theta) = \frac{1-E(\theta)}{2}$

we can compute,

$$E(\theta) \in [-1, +1]$$

~~$0 \leq f(\theta) \leq 1$~~ ~~to same~~

$$|0\rangle = e^{i \frac{1-E(\theta)}{2}} |0\rangle$$

$$E(\theta) = 1 - 2f(\theta)$$

lets say Phase oracle is $P_f(\theta) = e^{i f(\theta)} |0\rangle$

$$\Rightarrow P_f(\theta) = e^{i \frac{1-E(\theta)}{2}} |0\rangle$$

But we need ~~Phase~~ Phase oracle, $P_E(\theta) |0\rangle = e^{-i E(\theta) t} |0\rangle$

so, now we introduce a scaling factor $\rightarrow 2t$

$$\begin{aligned} P_{2tf} |0\rangle &= e^{i 2t \frac{1-E(\theta)}{2}} |0\rangle \\ &= e^{i t (1-E(\theta))} |0\rangle \\ &= e^{i t} \cdot e^{-i t E(\theta)} |0\rangle \\ &= e^{-i t E(\theta)} |0\rangle \end{aligned}$$

$\Rightarrow$ we can remove $e^{i t}$ as it is a global Phase & not acting on each parameter config.

now, we obtained our desired Phase oracle from probability oracle

$$U_f |0\rangle = \frac{1}{\sqrt{2}} (\sqrt{f(\theta)} |1\rangle + \sqrt{1-f(\theta)} |0\rangle) |0\rangle$$

$$P_{2tf} |0\rangle = e^{-i t E(\theta)} |0\rangle \Rightarrow \text{similar to Shor's eq } |4\rangle = e^{-i H t} |4\rangle$$

$\hookrightarrow$ time dependent Phase oracle / evolution operators for QHD

Amplification. APPROACH - 2.

Prob to find phase. (6)

$$H_0 = \sum_j c_j P_j \Rightarrow H_1 = c_j P_j$$

(LCU)

c_j = coefficient

P_j = Pauli strings $\rightarrow Z_0, X_0, X_1, Z_0, X_1, Z_2, \dots$

$$\alpha = \sum_j |c_j| \rightarrow \text{normalizing coefficient.}$$

$$H = \alpha \sum_j \frac{c_j}{\alpha} P_j$$

$$\frac{H}{\alpha} = \sum_j \frac{c_j}{\alpha} P_j$$

- LCU is unitary Has weighted sum of unitaries.
- $|0\rangle \rightarrow$ state Applying all the normalized coefficient weights.

- select applied corresponding Pauli/unitary based on index in $|0\rangle$
- block encoding is the result,
 H/α appears as on projected block of a larger unitary.

$\rightarrow$ separating sign & magnitude.

$$c_j = s_j |c_j|$$

$$s_j \rightarrow +0- \rightarrow c_j = +4, s_j = +1$$

$$|c_j| \rightarrow c_j = -4 \rightarrow 4$$

$$\frac{H}{\alpha} = \sum_j \frac{|c_j|}{\alpha} s_j P_j$$

$$\frac{H}{\alpha} = \sum_j a_j s_j P_j \Rightarrow \frac{|c_j|}{\alpha} = a_j$$

$\rightarrow$ Prepare the coefficient state

$$|a\rangle = \sum_j \sqrt{a_j} |j\rangle$$

$\rightarrow$ Prepare the select operation

$$\text{select } H = \sum_j |j\rangle\langle j| \otimes s_j P_j$$

$\rightarrow$ means apply the corresponding coefficients to $s_j P_j$

in a VQE problem, H is not diagonal so, we instead of computational basis we use eigen-basis of H , so we prepare
 $E(\theta) = \langle \psi(\theta) | H | \psi(\theta) \rangle$

$$\text{so } E(\theta) = \sum_j c_j \langle \psi(\theta) | H | \psi(\theta) \rangle$$

now compute

$$(\langle 01 | \otimes I) \text{selectH}(|01\rangle \otimes I)$$

now we have $|01\rangle = \sum_j \sqrt{a_j} |j\rangle$, substitute everything

$$(\langle 01 | \otimes I) \text{selectH}(|01\rangle \otimes I)$$

$$\Rightarrow \left(\sum_m \sqrt{a_m} \langle m | \otimes I \right) \left(\sum_j \sqrt{a_j} |j\rangle \otimes S_j^\dagger P_j \right) \left(\sum_k \sqrt{a_k} |k\rangle \otimes I \right)$$

263 gives

$$\sum_{j,k} \sqrt{a_k} (|j\rangle \langle j|k\rangle) \otimes S_j^\dagger P_j \quad j=k$$

$$\langle j|k\rangle = \begin{cases} 1 & \text{if } j=k \\ 0 & \text{if } j \neq k \end{cases} \quad \text{so they survive only if } j=k$$

$$\Rightarrow \sum_j \sqrt{a_j} |j\rangle \otimes S_j^\dagger P_j \quad \text{Kronecker delta}$$

now 164 gives

$$\left(\sum_m \sqrt{a_m} \langle m | \otimes I \right) \left(\sum_j \sqrt{a_j} |j\rangle \otimes S_j^\dagger P_j \right)$$

$$\sum_{m,j} \sqrt{a_m} \sqrt{a_j} \langle m | j \rangle \otimes S_j^\dagger P_j$$

$$\langle m | j \rangle = \begin{cases} 1 & , m=j \\ 0 & , m \neq j \end{cases} \Rightarrow \text{they work only when } j=m$$

$$\Rightarrow \sum_j a_j S_j^\dagger P_j$$

we know $a_j = \frac{|c_j|^2}{2}$, $S_j = \text{sign}(c_j)$

$$\Rightarrow \sum_j \frac{c_j}{2} P_j \Rightarrow \frac{H}{2}$$

$$\Rightarrow (\langle 01 | \otimes I) \text{selectH}(|01\rangle \otimes I) = \frac{H}{2}$$

Block Encoding

Let say $B = \frac{H}{2}$
 If H has energy E then B has its normalized version

$$B = \frac{E}{2}$$

$E = \lambda \alpha \rightarrow \lambda$ is the energy value given from B for one of its eigenstates.

From block encoding we have,

$$(\langle G | \otimes I) \text{ select } H (| G \rangle \otimes I) = \frac{H}{2} = B$$

Reflection operates with G being dealing with co-efficients.
 Reflection operates around $| G \rangle$

$$R_G = 2|G\rangle\langle G| - I \rightarrow \text{Builds a Reflection operator around } |G\rangle$$

Then walk operator is

$$W = (R_G \otimes I) \text{ select } H$$

Select $H \rightarrow$ Applied the signed Pauli terms

$R_G \otimes I \rightarrow$ Reflects the helper registers around H & coefficient state while leaving the system registers unchanged.

$\Rightarrow$ For each eigenvalue λ of $B \rightarrow \frac{\lambda}{2}$, W has phases $e^{\pm i \arccos(\lambda)}$

so W turns eigenvalue into rotation angle. walk operator

And then Applying a sequence of controlled phase rotations similar to QSP will get us $e^{-i\lambda t}$

We know $E = \lambda \alpha \Rightarrow \underline{e^{-iEt}}$

$\rightarrow$ Need to obtain this yet

unit¹, $B = \frac{H}{2}$, $E = \lambda x$, $\lambda = E/x$

①

After we have $(\langle G | \otimes I) \text{ select } H(\langle G | \otimes I) = \frac{H}{2} = B$,
 we use walk operators that turn the normalized H/x , which
 was only available as a projection block, into a unitary whose
 eigenphases encode the normalized eigenvalues of E/x .
 This gives us phase access to the Hamiltonian spectrum, which QSP then
 uses to construct phase oracle.

After walk operation, $W = e^{\pm i \arccos(E/x)}$

- walk operator acts like a rotation whose angle satisfies

$$\lambda = \cos(\theta) \Rightarrow \theta = \arccos(\lambda) = \arccos(E/x)$$

$$\begin{cases} W = e^{\pm i\theta} \\ \Rightarrow e^{\pm i \arccos(E/x)} \end{cases}$$

$$m = O(2 + \log(1/\epsilon)) \text{ qubits}$$

we do QSP phase synthesis,

to find all phases $\phi_0, \dots, \phi_m$

then build

$$U_\phi(w) = R_{\phi_0} W R_{\phi_1} W R_{\phi_2} \dots W R_{\phi_m} \approx e^{-iHt}$$

~~But this only works~~
 only when

$\phi = \phi_0, \dots, \phi_m$ are synthesized such that

$$f(x) = e^{-ixt} \quad x = \frac{E}{2} \quad \text{totally} \quad H \rightarrow B = \frac{H}{2} \text{ (normalized)}$$

$$\text{so, } f(E/2) = e^{-iEt}$$

$$UB|\psi\rangle = E/x |\psi\rangle$$

so, totally

$$|\psi\rangle \Rightarrow e^{-iEt} |\psi\rangle$$

W encodes $x = E/x$

ϕ chosen for $f(x) = e^{-ixt}$

$$U_\phi(w) = R_{\phi_0} W R_{\phi_1} W \dots W R_{\phi_m}$$

$$\Rightarrow U_\phi(w) \approx e^{-iHt}$$

QSP Phase Synthesis.

→ Walk operators gives $W = e^{+i\theta}$, $\theta = \arccos(\lambda)$, $\lambda = \frac{E}{2}$ (10)
 $\Rightarrow \cos(\theta) = \lambda = \frac{E}{2}$

→ e^{-iEt} for each eigenstate

$$e^{-iEt} = e^{-i\alpha x t} = e^{-iT x}, T = \alpha t$$

$$f(x) = e^{-ixT}$$

→ ~~Phase~~ Euler's formula,

$$e^{-iT x} = \cos(Tx) - i \sin(Tx)$$

By QSP, generated phases for $\cos(Tx)$ & $\sin(Tx)$ separately → Phase Synthesis.

$\Rightarrow \phi_{\cos}$ & $\phi_{\sin}$

we consider $\cos(Tx)$ to be $0.5 \cos(Tx)$ & same for $\sin$ for ~~stability~~ stability in QSP as $x \in [-1, 1]$

$$\text{so, } e^{-iT x} = 0.5 \cos(Tx) - i 0.5 \sin(Tx).$$

and for this we get $\phi_{\cos}$ & $\phi_{\sin}$.

Implementation Guide

(1)

Problem Hamiltonian H , $H = \sum_{j=0}^{j_0-1} c_j P_j$

$P \in \{I, X, Y, Z\}^{\otimes m}$

unitary
symmetries.

ex: $0.5 Z_0 - 1.2 X_0 X_1 + 0.7 Z_1 Z_2$

$\alpha = \sum_j |c_j| \Rightarrow H = \alpha \sum_{j=0}^{j_0-1} \frac{c_j}{\alpha} P_j$

$c_j = s_j |c_j|$, $s_j \in \{+1, -1\}$

$\Rightarrow H = \alpha \sum_{j=0}^{j_0-1} \frac{|c_j|}{\alpha} s_j P_j$

needed registers,

$|index\rangle |system\rangle |ancilla/work\rangle$

$|index\rangle \rightarrow$ stores which Pauli term P_j are we selecting
if we have L Pauli terms, we need, $m = \log_2 L$ ^{index} qubits

$|system\rangle \rightarrow$ The main register on which H acts on.

If H acts on n number of qubits, we need n ^{system} qubits.

$|ancilla/work\rangle \rightarrow$ This register helps in sign handling, uncomputing registers and some times in block encoding.

Prepare circuit, prepare $|0\rangle = \sum_j \sqrt{\frac{|c_j|}{\alpha}} |j\rangle$

- This circuit creates superposition over the Pauli indices ~~by the~~ weighted by ~~value~~ of normalized coefficient magnitudes.

$\rightarrow$ If one coefficient is max, it gets larger amplitude ~~value~~

After Prepare circuit,

$\rightarrow$ this only prepares coeff weights, not H .

$|0\rangle_{index} | \psi \rangle_{sys} \rightarrow \sum_j \sqrt{\frac{|c_j|}{\alpha}} |j\rangle_{index} | \psi \rangle_{sys}$

200k⁰ → Start.

← 20 → ~~1000~~ ~~network~~

→ Similar appearance.

→ Block Encoding: ~~not~~ the usual ~~block~~ Encoding.
~~→ hard~~ ~~→ hard~~

Q1

Given a Problem H , we can Prepare a Probability Oracle and a Phase Oracle. Now, if the H is too big and we are using Local Hamiltonians, then ~~how~~ practically how to combine the eigenphases of each local H . (mathematically we multiply them, means add the ~~phase~~ phases).

$$H = \sum_j C_j P_j \Rightarrow H = \sum_j C_j P_j \Rightarrow \frac{e^{-ixt}}{e^{-ix_1t} \dots e^{-ix_nt}} \Rightarrow$$

$H_1 \Rightarrow e^{-ix_1t}$	$U_1 \mid 0_1$	$0_1 \mid 0 \rangle = e^{-iE(0)t} \mid 0 \rangle$
$H_2 \Rightarrow e^{-ix_2t}$	$U_2 \mid 0_2$	$\vdots$
$\vdots$	$\vdots$	$\vdots$
$\vdots$	$\vdots$	$\vdots$

For multiple Local H 's over one ~~the~~ @ config, we have

$$0 \mid 0 \rangle = \underbrace{e^{-iE_1(0)t} e^{-iE_2(0)t} \dots}_{\text{while implementing new to combine these, as they are Prepared Separately.}} \mid 0 \rangle$$

while implementing new to combine these, as they are Prepared Separately.

Minimize energy function...

Local Min!

Given VAC now can we construct that over.
Simple $H \rightarrow VQE \rightarrow$ Phase orr.
 $H = \sum_i \sigma_i^x$ Bloch.

log function $\rightarrow$ over decoder -

Local Min!

Physical grounding

quantum chem (material sim.) $\rightarrow$ $\mu = \frac{1}{2}$

① $\uparrow \uparrow$ ionic model. TFIM.

$$H = -\sum_i z_i z_{i+1} - \sum_i \gamma_i$$

Phase $\rightarrow$ $Q(1)$ - Phase order $\rightarrow$ 1st

② -

Need to look into
circ computer
or architecture.

$(x, y) \rightarrow \text{Sample}$
 $x \rightarrow (x_1, x_2, x_3, \dots, x_n)$
 $y \in (0, 1)$

$$|0\rangle \rightarrow e^{-i\epsilon(0)}|0\rangle$$

$U_{\text{vac}} \rightarrow U_{\text{vac}}(\theta)$
 $\theta \rightarrow \text{Rotation angles / Parameters}$

$M \rightarrow \text{several local terms.}$

Amplitude enc -

$$|\Theta\rangle = \sum_K a_K |\theta^K\rangle$$

1st block

$$\rightarrow |\psi\rangle = \sum_K a_K |\theta^K\rangle |x\rangle |y\rangle |0\rangle |0\rangle$$

U_{vac}
 $\hookrightarrow U_{\text{vac}}(\theta^K)$

$$\hookrightarrow \text{o/p} \rightarrow \sqrt{1-P_K} |0\rangle + \sqrt{P_K} |1\rangle$$

$$P_K = P_{\theta^K}(x)$$

$$\Rightarrow \sum_K a_K |\theta^K\rangle |P_K(x)\rangle |y\rangle (\sqrt{1-P_K} |0\rangle + \sqrt{P_K} |1\rangle) |0\rangle$$

USE XOR for ~~xxx~~ computation

$$c = 0 \oplus x$$

After VAC + XOR



2

→ $U_1 |0\rangle |0\rangle = |0\rangle (\sqrt{1/2} |1\rangle |\phi_1(\theta)\rangle + \sqrt{1/2} |0\rangle |\phi_0(\theta)\rangle)$

→ ~~Prob~~ Probability Oracle

→ Prob Oracle

$U_1 \Rightarrow O_1$

→ Phase Oracle

→ Prob Oracle & Phase Oracle.

→ program 3.9 in efficient quantum algorithms for quantum optimal control.

$e^{-i l(\theta^k; x, y)} |\theta^k\rangle$

→ $\sum_K a_K e^{-i l(\theta^k; x, y)} |\theta^k\rangle |0\rangle$
 helps bits.

For phase oracle → Additive.

Phases into binary tree.

output
some
approx
by
fine.

★ From Heisenberg-ly.

3 Conclusion and future research

We gave a new approach to quantum gradient computation that is asymptotically optimal (up to logarithmic factors) for a class of smooth functions. In terms of the number of queries needed to estimate the gradient within fixed error with respect to the maximum, this is best known result. Our approach includes the use of differentiation formulas originating from high-order Taylor series expansions. These high-order methods quadratically improve the scaling of

$|\psi_0(x)\rangle \rightarrow$ variational state

$x \rightarrow x_1, x_2, x_3, \dots \rightarrow$ optimization parameters

$$\frac{1}{\sqrt{2}}|0\rangle + \frac{1}{\sqrt{2}}|1\rangle \rightarrow |\psi\rangle$$

$$\Rightarrow \frac{1}{2} (1) + \frac{1}{2} (3) \rightarrow \langle \psi | H | \psi \rangle$$

$$H = \begin{bmatrix} 1 & 0 \\ 0 & 3 \end{bmatrix}$$

$$|\psi(x)\rangle = \frac{\sqrt{2}}{2} |0\rangle + \frac{1}{2} |1\rangle$$

$$= \frac{3}{4} (1) + \frac{1}{4} (3)$$

$$= \frac{3}{4} + \frac{3}{4} = 1.5$$

$$\langle \psi(x) | H | \psi(x) \rangle \rightarrow E \rightarrow \text{Energy}$$

$$\frac{1}{\sqrt{N}} \sum_{i=0}^{N-1}$$

$$H = \begin{bmatrix} 1 & 0 \\ 0 & 3 \end{bmatrix}$$

$$\langle \psi(x) | H | \psi(x) \rangle$$

$$\langle \psi(x) | H | \psi(x) \rangle$$

(1)

$$\langle \psi(x) | H | \psi(x) \rangle$$

using quantum formula.

$$E(s) = \sum_i h_i s_i + \sum_{i,j} J_{ij} s_i s_j + \theta$$

where $s \in \{-1, +1\}$

→ combination equation for kinetic & potential energy

Dragnetman Lagrangian.

$$L(x, \dot{x}, t) = e^{\alpha(t)} \left(\frac{1}{2} \|\dot{x}\|^2 - f(x) \right)$$

($\rightarrow$) $f(x) \rightarrow$ loss function

$e^{\alpha(t)}$ → damping effect to prevent bouncing of gradient.

$\frac{dL}{dt} = \frac{d}{dt} L(x, \dot{x}, t)$ Dragnetman Lagrangian → different x just symbolic

$$H(t) = e^{\alpha(t)} \left(\underbrace{-\frac{1}{2} \dot{x}^2}_{\text{kinetic energy term}} + \underbrace{f(x)}_{\text{potential energy term}} \right)$$

~~$H(t) = e^{-\alpha(t)}$~~

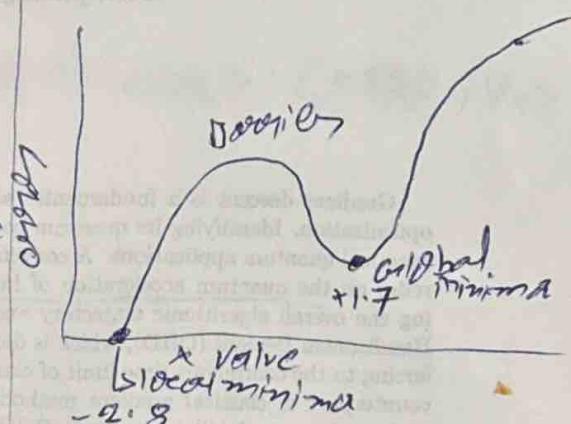
$$\psi(t) = e^{-\alpha(t)} \psi(0)$$

Let's say,

we have a parameter x
 $x \in (-3, +3)$
 for instance
 " " on Loss function $f(x)$

→ we need $x^* = \arg \min f(x)$

gradient decent → classical
 $x = -2.8$ (Random initialization)
 total wave function,



calculate gradient.

$$x_{new} = x_{old} - \eta \frac{\partial f}{\partial x}$$

$\frac{\partial f}{\partial x} =$ gradient
 η will be negative going down
 η learning rate

→ stops at some where -1.96 (local minimum)

as $\frac{\partial f}{\partial x}$ becomes 0
 so, $x_{new} = x_{old} - 0 \times \eta$
 $\Rightarrow x_{new} = x_{old}$

⇒ issue with non-convex optimization.

classical optimization issue

Implementation

Given,

$$E_k = \langle \psi_k | H | \psi_k \rangle$$

In superposition: $\sum a_k |k\rangle$

$$\Rightarrow \sum_k a_k |k\rangle \rightarrow \sum_k a_k e^{-iE_k t} |k\rangle$$

$$e^{-iE_k t} |k\rangle$$

k -label of candidate solution set

E_k - objective value.

$t \rightarrow$ time evolution parameters.

Hamiltonian.

$$H = \sum_{j=0}^{L-1} c_j P_j \quad \text{if local } H \text{ is used.}$$

normalize, $\tilde{H} = \frac{H}{\alpha} \quad H|k\rangle = E_k |k\rangle \Rightarrow H|k\rangle = E_k \alpha |k\rangle$

for every candidate, $x_k = \frac{E_k}{\alpha}$

$a_j = \frac{|c_j|}{\alpha}$, state coefficient state $|a\rangle$

$$|a\rangle = \sum_{j=0}^{L-1} \frac{|c_j|}{\alpha} |j\rangle = \sum_{j=0}^{L-1} \sqrt{\frac{|c_j|}{\alpha}} |j\rangle$$

this is prepared

now, we select operator,

$$S_j = \text{sign}(c_j) \quad S_j = \begin{cases} +1, & c_j \geq 0 \\ -1, & c_j < 0 \end{cases}$$

$$\text{select} = \sum_{j=0}^{L-1} |j\rangle \langle j| \otimes S_j P_j$$

Probability Oracle.

$$P_k = \frac{1-x_k}{2} \Rightarrow \frac{1}{2} (1 - \frac{E_k}{\alpha})$$

After Applying i^{th} Hadamard gate,

$$H_A |0\rangle_A = \frac{|0\rangle_A + |1\rangle_A}{\sqrt{2}} \Rightarrow \text{total state} \Rightarrow \frac{1}{\sqrt{2}} |k\rangle_K (|0\rangle_A |x_k\rangle + |1\rangle_A |x_k\rangle)$$

now controlled select.

$$C_A(\text{select}) = |0\rangle\langle 0|_A \otimes I + |1\rangle\langle 1|_A \otimes \text{select}$$

$$\Rightarrow \frac{1}{\sqrt{2}} |K\rangle (|0\rangle_A |x_K\rangle + |1\rangle_A \text{select} |x_K\rangle)$$

After 2nd Hadamard gate.

$$\Rightarrow \frac{1}{2} |K\rangle_K |0\rangle_A (I - \text{select}) |x_K\rangle + |1\rangle_A (I - \text{select}) |x_K\rangle$$

normalized state on ancilla is, $\frac{1}{2} (I - \text{select}) |x_K\rangle$

$$P_K = \frac{1}{2} \langle x_K | (I - \text{select})^\dagger (I - \text{select}) |x_K\rangle$$

and unitary, $\text{select}^\dagger = \text{select}$

$$\Rightarrow (I - \text{select})^\dagger (I - \text{select}) = (I - \text{select})^2 = I - 2\text{select} + \text{select}^2$$

$$= 2I - 2\text{select}$$

$$\Rightarrow P_K = \frac{1}{4} \langle x_K | (2I - 2\text{select}) |x_K\rangle = \frac{1}{2} [1 - \langle x_K | \text{select} |x_K\rangle]$$

$$\text{so, } P_K = \frac{1}{2} \left(1 - \frac{E_K}{\alpha}\right) \text{ as } \langle G | \otimes I \rangle \text{select} (|G\rangle \otimes I) = \frac{H}{2}$$

$$\Rightarrow \frac{\langle \Psi_K | H | \Psi_K \rangle}{\alpha}$$

In terms of implementation,

$$U_P |K\rangle |0\rangle_{\text{aux}} = |K\rangle (\sqrt{P_K} |\Psi_{K,\text{good}}\rangle |1\rangle_A + \sqrt{1-P_K} |\Psi_{K,\text{bad}}\rangle |0\rangle_A)$$

now, E_K is in form of P_K .

using geometry of the states,

$$U_P |K\rangle |0\rangle_{\text{aux}} = |K\rangle (\sin \theta_K |\Psi_{K,\text{good}}\rangle + \cos \theta_K |\Psi_{K,\text{bad}}\rangle)$$

$$= |K\rangle (\sin \theta_K |\Psi_{K,\text{good}}\rangle |1\rangle + \cos \theta_K |\Psi_{K,\text{bad}}\rangle |0\rangle)$$

by constructing geometry reflection operators,

the probabilities are encoded through amplitudes $\sin \theta, \cos \theta$

so from U_P we can create a unitary whose eigen values are $e^{\pm 2i\theta}$

$$R_0 = 2I_K \otimes |0\rangle\langle 0| - I \rightarrow \text{Reflection by bad}$$

$$R_1 = 2I_K \otimes |1\rangle\langle 1| - I \rightarrow \text{Reflection by good state}$$

Now the grover operator,

$$G = R_0 U_P^\dagger R_1 U_P \Rightarrow (2\pi I - I) U_P^\dagger (2\pi I - I) U_P$$

G had 2 eigen vectors in the subspace

$$|\phi_+\rangle, |\phi_-\rangle$$

$$\Rightarrow G|\phi_{k\pm}\rangle = e^{\pm 2i\theta_k} |\phi_{k\pm}\rangle, G|\phi_{k-}\rangle = e^{-2i\theta_k} |\phi_{k-}\rangle$$

we know, $P_k = \sin^2 \theta_k \Rightarrow \cos(2\theta) = 1 - 2\sin^2 \theta = 1 - 2P_k$

$$P_k = \frac{1}{2} \left(1 - \frac{E_k}{\alpha} \right)$$

$$\cos 2\theta = 1 - 2 \left(\frac{1}{2} \left(1 - \frac{E_k}{\alpha} \right) \right) = 1 - \left(1 - \frac{E_k}{\alpha} \right) = \frac{E_k}{\alpha}$$

$$E_k = \alpha \cos 2\theta$$

~~G~~ involves operation folds π - in superposition.

And $\cos$ is an even function and $\cos(-\theta) = \cos \theta$

so $G^\dagger F_r(\phi) = e^{-i\tau \cos \phi}, \tau = \alpha t$
 $= e^{-i\tau \alpha \cos \phi} = e^{-i\tau E_k}$

~~so $G^\dagger F_r(\phi) | \phi \rangle$~~

$$F_r(\alpha) | \phi_{k\pm} \rangle | 0 \rangle = e^{-i\tau E_k} | \phi_{k\pm} \rangle | 0 \rangle$$

one question.

why do we need Fourier transformation. A little unclear.

$$m = -m, -m+1, \dots, m, F_m(\phi) = \sum_m P_m e^{im\phi}$$

If m is large enough then, $F_m(\phi) \approx F_r(\phi)$

$$\Rightarrow F_m(\phi) \approx e^{-i\tau E_k}$$

why Fourier? $\Rightarrow$ id, $G^2 |\phi_{k\pm}\rangle = G(e^{\pm i\theta_k} |\phi_{k\pm}\rangle)$
 $= e^{\pm i\theta_k} G |\phi_{k\pm}\rangle$
 $= e^{\pm i\theta_k} (e^{\pm i\theta_k} |\phi_{k\pm}\rangle)$
 $= e^{\pm 2i\theta_k} |\phi_{k\pm}\rangle$

to implement, $\sum_m P_m G^m$

we use LCU

Full LCU is

$$A_{LCU} = (P_0 P_1 \dots P_{\beta-1}^\dagger \otimes I) \text{ select } G \text{ (phase } P \text{ prepare } P^\dagger \otimes I)$$

Starts with $|0\rangle_m |\phi\rangle \rightarrow$ After $P_0 \dots P_{\beta-1} = \frac{1}{\sqrt{\beta}} \sum_m \sqrt{\frac{|P_m|}{\beta}} |m\rangle |p\rangle$

After phase, $\sum_m \sqrt{\frac{|P_m|}{\beta}} e^{i\theta_m} |m\rangle |p\rangle$

After select, $\sum_m \frac{\sqrt{|\beta_m|}}{\sqrt{N}} e^{i\delta_m} |m\rangle G^m |\psi\rangle$

we get finally,

$$\frac{1}{\lambda} \sum_{m=-M}^M \beta_m G^m |\psi\rangle$$

we then use OAA - *oblivious Amplitude amplification*
 → not all branches with power registers
 not returned to zero.

$$A_{LU} |0\rangle_m |\psi\rangle = |0\rangle_m B |\psi\rangle + |\psi\rangle$$

OAA increases the amplitude of the success sub space and acts independently on unknown system state $|\psi\rangle$

$$\Pi = |0\rangle\langle 0|_m \otimes I \text{ success projector}$$

branches all work registers acting on G

$$R_\Pi = 2\Pi - I$$

$$B_{OAA} = (|0\rangle\langle 0|_m \otimes I) A_{OAA} (|0\rangle\langle 0|_m \otimes I)$$

$$A_{OAA} = A_{LU} R_\Pi A_{LU}^\dagger R_\Pi A_{LU}$$

$$B_{OAA} = \pm (3B - 4B^3) \rightarrow \text{successes of the sequence}$$

~~not sure~~
 $B \propto SU, S = 1/\lambda_B \Rightarrow B^\dagger B = S^2 I$

$$\Rightarrow 3B - 4B^3 = 3SU - 4(SU)(SU)^\dagger (SU) = 3SU - 4S^3 U U^\dagger U$$

$$= 3SU - 4S^3 U = (3S - 4S^3)U \Rightarrow \left| \frac{3}{\lambda_B} - \frac{4}{\lambda_B^3} \right| = S_{OAA}$$

$$B_{OAA} \approx \pm S_{OAA} U \approx F_T(\omega) \approx e^{-iE_k t}$$

$$B_{OAA} |\phi_{k,\pm}\rangle \approx \pm S_{OAA} e^{iE_k t} |\phi_{k,\pm}\rangle$$

Approximating $3S - 4S^3 \rightarrow S_{OAA}$ to one $\rightarrow 1$ will give us

$$B_{OAA} |\phi_{k,\pm}\rangle \approx e^{-iE_k t} |\phi_{k,\pm}\rangle$$

Preparing β coefficients.

target function is $f(\phi) = e^{-i\tau \cos \phi}$ $f_m(\phi) = \beta_{-2} e^{-2i\phi} \dots \beta_2 e^{2i\phi}$
 we get through $\beta_m = \frac{1}{2\pi} \int_0^{2\pi} e^{-i\tau \cos \phi} e^{-im\phi} d\phi$ exponent with power of m

$\rightarrow F_m(\phi) = \sum_{n=-m}^m \beta_n \phi^n$ $\hookrightarrow$ converges unitary over powers.

$\lambda = \sum_m |\beta_m| \Rightarrow |\beta\rangle = \sum_m \frac{|\beta_m|}{\lambda} |m\rangle$ $\left\{ \begin{array}{l} \text{coefficients} \\ \text{of Powers} \end{array} \right.$

$\text{Phase } \beta = \sum_m \frac{\beta_m}{|\beta_m|} |m\rangle \langle m|$

combine this with $\langle \nu | \rightarrow \frac{1}{\lambda \beta} \sum_m \beta_m \phi_m \frac{1}{\lambda \beta} e^{-i\tau \cos \phi}$

$|0\rangle \rightarrow |\beta\rangle$

Prepare β state ϕ Phase β

$\beta_m = |\beta_m| e^{i\theta_m}$ As β is a complex number.
 $\hookrightarrow$ magnitude $\hookrightarrow$ phase

$|\beta\rangle = \sum_m \frac{|\beta_m|}{\lambda_\beta} |m\rangle$ where $\lambda_\beta = \sum_m |\beta_m|$

phase $\beta \Rightarrow \frac{\beta_m}{|\beta_m|} = e^{i\arg(\beta_m)}$

$\hookrightarrow |m\rangle = e^{i\arg(\beta_m)} |m\rangle \langle m|$

Prepare β encodes how big the ~~coefficient~~ coefficient is and
 phase β encodes is complex angle.